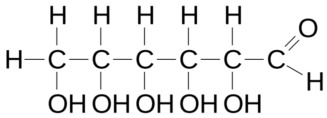
2009 9746 H2 Chemistry

Paper 1 Answer Key:

1	A	11	C	21	A	31	A
2	B	12	B	22	D	32	D
3	A	13	C	23	D	33	D
4	A	14	B	24	D	34	B
5	C	15	D	25	B	35	D
6	B	16	C	26	A	36	D
7	D	17	B	27	C	37	A
8	C	18	D	28	C	38	B
9	B	19	B	29	C	39	D
10	A	20	D	30	B	40	C

Paper 1 Worked Solutions

Q1 Answer: A



ROH + Na → RO^-Na^+ + ½ H_2
Since there are 5 mol of alcohol groups, 5 mol of sodium will react to form 2.5 mol of hydrogen gas.

Q2 Answer: B

Amt of $C_2O_4^{2-}$ = 2×10^{-3} mol
$KHC_2O_4.H_2C_2O_4 \equiv 2C_2O_4^{2-}$

Amt of MnO_4^- = $\frac{2.0\times10^{-3}}{5}\times 2$ = 8×10^{-4} mol

Vol of MnO_4^- = $\frac{8.0x10^{-4}}{0.02}$ = 0.04 dm^3 = 40 cm^3

Q3 Answer: A
Only cations are reduced at cathode.

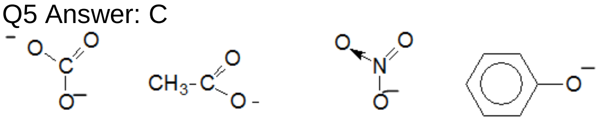
Based on the proton number and electronic configuration, options B, C and D are anions.

Q4 Answer: A

Al to Al^{2+} involves 1st IE + 2nd IE = 577 + 1820
= 2397 kJ mol^{-1}

1st IE of Co + 2nd IE of Co^{2+} = 757 + 1649
= 2397 kJ mol^{-1}

Q5 Answer: C



CO_3^{2-} CH_3COO^- NO_3^- phenoxide

All have delocalised electrons.

Q6 Answer: B

"Dry" means absence of water. Hence dry HCl remains as a simple molecule. There are no ions, hence it cannot conduct electricity.

Q7 Answer: D
pV = nRT

p = $\frac{(1.6x10^{-3})(8.31)(273+273)}{3x10^{-3}}$

*change temperature to Kelvin and volume to m^3

Q8 Answer: C

$|\Delta H_{hyd}|\ \alpha\ |\frac{q^+}{r^+}|$

Since Ca^{2+} has larger ionic radius than Mg^{2+}, Ca^{2+} has lower $|\Delta H_{hyd}|$ than Mg^{2+}.

Q9 Answer: B
'Process works by applying a constant pressure' shows that reaction is non-spontaneous. ΔG is positive.

After separation, the system is now more orderly as there are only water molecules present.

Q10 Answer: A
When water freezes, it is at equilibrium (ice and water in equilibrium). ΔG = 0

ΔG = ΔH – TΔS
0 = 6 – (273)ΔS
ΔS = -21.97 J mol^{-1} K^{-1}

Amt of water = 54 / 18 = 3 mol

ΔS = -21.97 X 3 ≈ - 66 J K^{-1}

Q11 Answer: C

$E(Zn^{2+}/Zn) = -0.76$ V (oxd)
$E(H^+/H_2) = 0$ (red)

Reaction occurring: $2H^+ + 2e \rightarrow H_2$
$Zn \rightarrow Zn^{2+} + 2e$
$E_{cell} = E_{red} - E_{oxd}$
$= 0 + 0.76$

If it is > 0.76, oxidation is favoured or reduction was favoured

$[Zn^{2+}]$ cannot be greater than 1 mol dm^{-3} as it will cause the E_{cell} to decrease.
$[H^+]$ can be greater than 1 mol dm^{-3} as it will facilitate reduction (by LCP) and E_{red} will become more positive
➔ E_{cell} becomes more positive

Q12 Answer: B

Amount (pressure) of gaseous particles increases as dissociation of X_2 is favoured at higher temperature (bond dissociation is endothermic). Gradient of graph doubles when X_2 dissociates to X atoms at higher temperature

Q13 Answer: C
At lower [ethanal], active sites of enzymes are not fully occupied, thus reaction is 1^{st}-order with respect to ethanal.
At higher [ethanal], all active sites are occupied and reaction is zero order with respect to ethanol.

Q14 Answer: B
Water vapour is produced when aluminium hydroxide decomposes, the water vapour will then extinguish the flames.

Q15 Answer: D
Cu: $[Ar]3d^{10}4s^1$ Cu^{2+}: $[Ar]3d^9$
Mg: $[Ne]3s^2$ Mg^{2+}: [Ne]

Cu has 2 more electron shells than Mg^{2+}

Q16 Answer: C
$3H_2O_2 + Cr_2O_7^{2-} + 8H^+ \rightarrow 2Cr^{3+} + 7H_2O + 3O_2$
(-1) (0)

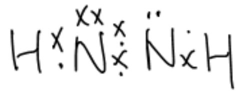
$H_2O_2 + 2I^- + 2H^+ \rightarrow H_2O + I_2$
(-1) (–2)

Q17 Answer: B
1^{st} IE of elements 1 – 3 are comparable ➔ likely to be transition elements. Referring to Data Booklet,

	Co	Ni	Cu	Zn
1^{st} IE/kJmol^{-1}	757	736	745	908

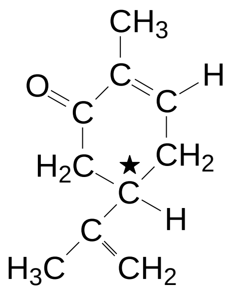
Q18 Answer: D
$2Cu^{2+}(aq) + 4I^-(aq) \rightarrow 2CuI(s) + I_2(aq)$
White ppt brown solution
$2S_2O_3^{2-} + I_2 \rightarrow S_4O_6^{2-} + 2I^-$
$E^\ominus_{cell} = +0.15V - (+0.54V) = -0.39V$

Q19 Answer: B

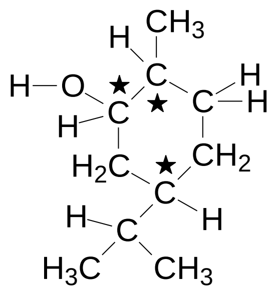


There are 2 bond pairs and 1 lone pair around N.
Shape: bent bond angle: <120°

Q20 Answer: D



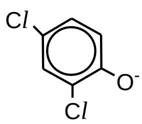
Carvone



after reduction

Q21 Answer: A
This is an electrophilic addition reaction where bromine is the electrophile.

Q22 Answer: D
This is a nucleophilic substitution reaction where

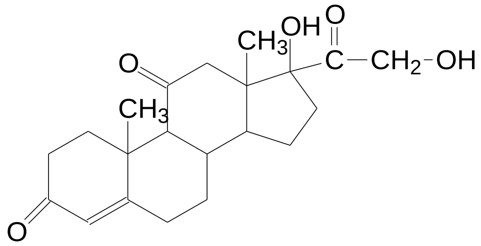


is the nucleophile.

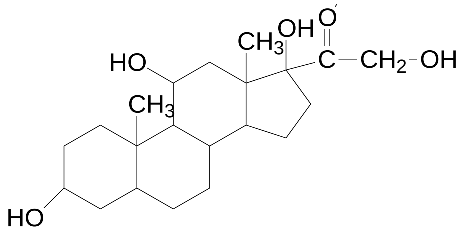
Q23 Answer: D
'No precipitate seen when boiled with ethanolic silver nitrate' → halogenoarene

Q24 Answer: D
2 bromine atoms are added across the C=C bond. 2 bromine atoms are substituted at positions 2 and 4 with respect to phenol. Total of 4 bromine atoms added

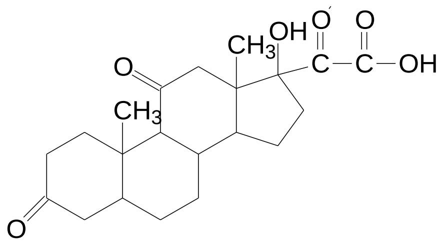
Q25 Answer: B



cortisone



After reduction



After oxidation

There are four double bonds in final product.

Q26 Answer: A

Note that $NaBH_4$ can only reduce carbonyl compounds, not the C=C bond.

$$CH_2{=}CHCH{=}O \xrightarrow{NaBH_4} \underset{(P)}{CH_2{=}CHCH_2OH} \xrightarrow{K_2Cr_2O_7/H^+} \underset{(Q)}{CH_2{=}CHCOOH}$$

Q27 Answer: C

Phenol is the least acidic as The p-orbitals of oxygen overlaps with the π electron cloud of the benzene ring This results in the delocalisation of negative charge on the oxygen into the ring which reduces the intensity of the negative charge on the oxygen atom. Phenoxide ion is stabilised by charge dispersal

Chlorine is a stronger electron-withdrawing group than bromine, thus chloroethanoic acid is the most acidic of the 4 compounds.

Q28 Answer: C

Acid will react with amine to form a salt which is soluble. Benzene will not react with acid.

Q29 Answer: C

Denaturation takes place with insulin is heated with 6 mol dm^{-3} HCl → amide linkages are broken.
The amino acids formed then undergo neutralisation.

Q30 Answer: B

4 moles of water are removed.
Total M_r = 2(75) + 89 + 2(105) – 4(18) = 377

Q31 Answer: A

Empirical formula: C_2H_4O or C_4H_8S (option 3)
Molecular formula: $C_4H_8O_2$ (options 1 and 2)

Q32 Answer: D

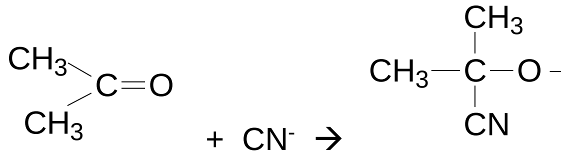
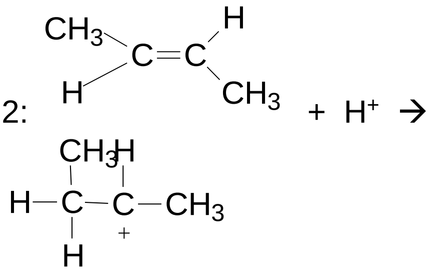
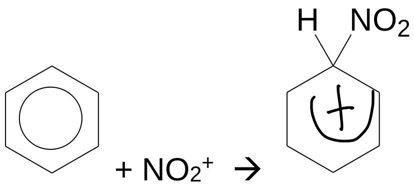
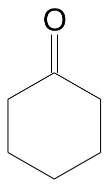
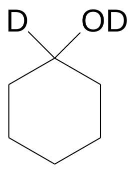
1: C–C–C bond angle in diamond is 109.5° and C–C–C bond angle in graphite is 120°. Hence option 1 is correct.
2: The shortest C-C bond is in graphite due to sp^2-sp^2 overlap.
3: All covalent bonds in diamond and graphite are of same strength.

Q33 Answer: D

1: As [ethanol] increases, position of equilibrium shifts to the right, favouring oxidation, hence E^o becomes more negative.
2: Oxygen is reduced at electrode 2.

Q34 Answer: B

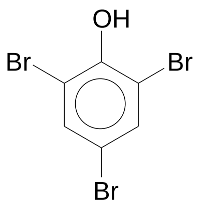
1: From the table, the half-life of 2.0 x 10^{-4} s can be determined.
2: With a constant half-life, the reaction is first order wrt to OH^-.
3: There is not enough information given in the question to determine the order of reaction wrt methane.

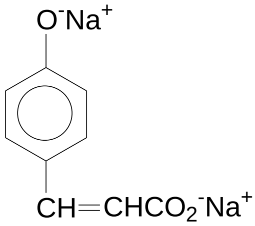
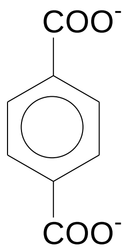
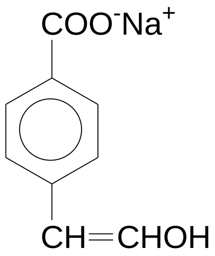
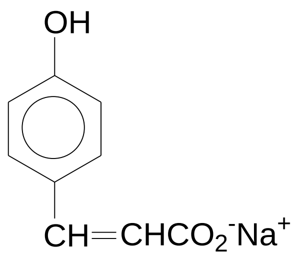
Q35 Answer: D 1: This can be determined from data booklet. 2: Since $Sr(OH)_2$ is more soluble, it will not precipitate better. 3: There is no metallic bonding in hydroxyapatite	Q36 Answer: D 1: It is an oxidising agent in reaction 2, where it is reduced to Ag. 2: Ammonia forms a soluble complex with silver ions. 3: Fe^{2+} ions are reagents in reaction 2.
Q37 Answer: A 1:  2:  3: 	Q38 Answer: B 1: $CH_3C{\equiv}N + 2D_2 \rightarrow CH_3CD_2ND_2$ 2:  can be reduced to  using D_2 and Ni catalyst 3: cannot be formed
Q39 Answer: D 1: acyl chloride reacts with alcohol to form ester. 2: the carboxylic acid can react further with the tertiary amine to form a salt. 3: acyl chloride cannot react with carboxylic acid.	Q40 Answer: C 1: the isoelectronic point may not be at pH 7. 2: at half equivalence point, there are equal amounts of $H_3N^+CHRCO_2H$ and $H_3N^+CHRCO_2^-$. 3: At equivalence point, $H_3N^+CHRCO_2^-$ is present.

Paper 2 Worked Solutions

1(a)	Lattice energy is the amount of energy released when 1 mole of the solid ionic MgO is form from its constituent gaseous Mg^{2+} and O^{2-} ions at standard conditions. It is an exothermic process Mg^{2+} (g) + O^{2-} (g) → MgO (s)
1(b)(i)	Based on the lattice energy equation, where $\left\|\Delta H_{latt}\right\| \alpha \left\|\frac{q^+q^-}{r^+ + r^-}\right\|$, ionic radius: Cl^- (0.181nm) < Br^- (0.195nm)< I^- (0.216nm) charge of all the ions remains unchanged. Lattice energy: NaCl > NaBr > NaI *Note: trend must be shown. Otherwise, marks will be penalized.* *Note: ionic radii of all 3 halide ions must be quoted.
1(b)(ii)	Using the $\left\|\Delta H_{latt}\right\| \alpha \left\|\frac{q^+q^-}{r^+ + r^-}\right\|$, when comparing the cations, ionic radius: Na^+ > Mg^{2+} ionic charge: Na^+ < Mg^{2+} when comparing the anions, ionic radius: X^- > O^{2-} ionic charge: X^- < O^{2-} Electrostatic forces of attraction: MgO > NaX Energy required to overcome: MgO > NaX Lattice energy: MgO > NaX *Note: Alternatively: [1] for comparison of all ionic radius of ions and [1] for comparision of all ionic charge.
1(c)	Iodide has a large ionic size, and thus a large anionic electron cloud. Therefore, the electron cloud can be polarized by the Ag^+ ion, thus inducing some covalent character into the ionic bond. This results in the difference in the experimental and theoretical values.
1(d)(i)	When used to refer to a crystal lattice, co-ordination number refers to the number of ions that surround another ion of opposite charge.
1(d)(ii)	Cs^+ is much larger as compared to the Na^+ and K^+ ion. Thus, this allows more Cl^- ions to be able to be packed around the Cs^+ ion.

2(a)	Zn (metal 3) and Cu (metal 4)
2(b)	The residue is Cu. Cu^{2+}/Cu E^{O} value is +0.34 V, and compared to the H^+/H_2 value of 0.00 V, it will not be able to undergo oxidation in hydrochloric acid. Thus, it will not dissolve. $E^{O}_{cell} = 0 - (0.34) = -0.34V$ (reaction not feasible) *make reference to E^{O} values
2(c)(i)	Sulfuric acid behaves as an oxidizing agent.
2(c)(ii)	$SO_4^{2-} + Zn + 4H^+ \rightarrow SO_2 + 2\,H_2O + Zn^{2+}$
2(d)(i)	$Al(OH)_3$
2(d)(ii)	$Al^{3+} + 3\,OH^- \rightarrow Al(OH)_3$ $Al(OH)_3 + OH^- \rightarrow Al(OH)_4^-$ *penalize if NaOH is written, since qn required ionic equation.
2(e)	Zinc has an electronic configuration of a fully filled outer 4s orbital. Thus, it shows some properties of Group II elements
2(f)(i)	$Zn(NO_3)_2 \rightarrow ZnO + 2\,NO_2 + ½\,O_2$
2(f)(ii)	Ionic radius: Mg^{2+} (0.065 nm) < Zn^{2+} Mg^{2+} has a greater polarizing power and charge density than Zn^{2+} Thus, Zn^{2+} has smaller polarizing effect on the anion NO_3^- charge cloud. Thus, $Zn(NO_3)_2$ will decompose at a higher temperature. *penalise [1] if no reference to Data Booklet values
2(g)	$Zn^{2+} + 2OH^- \rightleftharpoons Zn(OH)_2$ or $Zn^{2+} + 2OH^- \rightarrow Zn(OH)_2$ $Zn(OH)_2 + 4\,NH_3 \rightleftharpoons Zn(NH_3)_4^{2+} + 2\,OH^-$ or $Zn(OH)_2 + 4\,NH_3 \rightarrow Zn(NH_3)_4^{2+} + 2\,OH^-$ * $Zn(OH)_2$ must appear in second equation

3(a)	Ethanoic acid and benzoic acid
3(b)	$\left[H^+\right]_{phenol} = \sqrt{K_a \times c}$ $[H^+]_{phenol} = 3.606 \times 10^{-6}$ mol dm^{-3} pH = **<u>5.4</u>**
3(c)	Phenol is used as a weak acid as the negative charge of the phenoxide ion can be delocalized into the delocalized pi electron cloud of the benzene ring, thus stabilizing the conjugate base, hence releasing the H^+ ion Methanol is a even weaker acid as the negative charge of the methoxide ion is intensified due to the presence of the electron donating methyl group, thus destabilizing the conjugate base.
3(d)	Decolourisation of reddish brown Br_2 (aq) and white precipitate of 2,4,6-tribromophenol is seen.  *Must be 2,4,6-tribromophenol
3(e)(i)	Esterification process using methanol in the presence of concentrated sulfuric acid and reflux
3(e)(ii)	Step 1: nucleophilic substitution to convert ethanoic acid to ethanoyl chloride using PCl_5 at room temperature Step 2: Formation of ester using phenol with ethanoyl chloride at room temperature

4(a)(i)	A, C, D
4(a)(ii)	A, D
4(a)(iii)	A, D
4(a)(iv)	A, D
4(a)(v)	D
4(b)(i)	Reaction (v)
4(b)(ii)	
4(c)	
4(d)	A produces  D produces 

5(a)(i)		C	H	O	
	% composition	70.6	5.9	23.5	
	÷ A_r	5.883	5.9	1.469	
	÷ 1.469	4	4	1	

Empirical formula of **E**: <u>**C_4H_4O**</u>

5(a)(ii)

Using pV = nRT

$$M_r = \frac{0.344 \times 8.314 \times (21+273)}{(101.7\times10^3)\times(60.4\times10^{-6})} = \underline{\mathbf{136.8}}$$

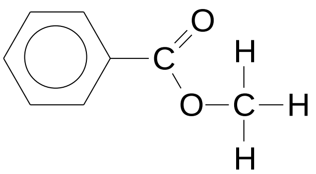
Let molecular formula be $(C_4H_4O)_n$

n(4(12) + 4(1) + 16) = 136.8

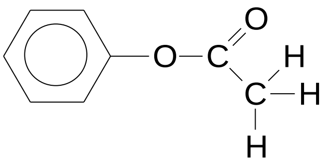
<u>**n = 2**</u>

Molecular formula of **E**: <u>**$C_8H_8O_2$**</u>

5(b)(i)

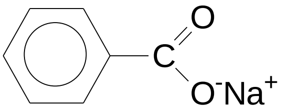
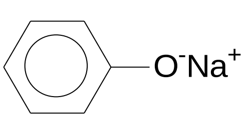
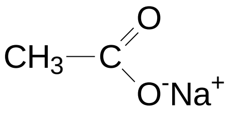
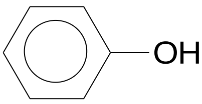
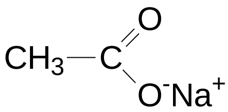
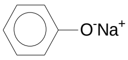


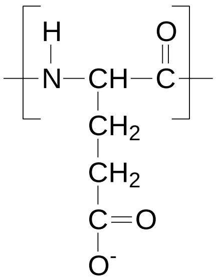
E1:



E2:

*Note: Displayed formula!!!

5(b)(ii)	For methyl benzoate (**E1**),  and CH_3OH For phenyl ethanoate (**E2**),  and 
5(c)	For **E1**, no reaction For **E2**,  and  Carbonic acid is a stronger acid than phenol. This is indicated by the K_a values given in question 3. Thus, carbonic acid will react with  to form phenol.

1(a)	**Primary structure** of proteins refers to the **<u>sequence of amino acid residues</u>** in the polypeptide in which the amino acids are joined together by **<u>peptide bonds</u>**. **Secondary structure** of proteins describes the **<u>regular folding</u>** of the **<u>polypeptide backbone</u>** into structural pattern of **<u>α-helix</u>** or **<u>β-pleated sheets</u>** due to intramolecular or intermolecular **<u>hydrogen bonds</u>** formed between the **<u>C=O$^{\delta-}$ of a peptide bond</u>** and **<u>N-H$^{\delta+}$ of another peptide bond</u>**. **Tertiary structure** of protein describes the **<u>overall folding of the secondary structures/polypeptide chains</u>** into a **<u>3-dimensional compact shape</u>** due to **<u>ionic bonds</u>**, **<u>hydrogen bonds</u>**, **<u>Van der Waals' forces</u>** and **<u>disulphide bonds</u>** between the **R groups (side chains)** of the polypeptide. **Quaternary structure** of protein describes how different tertiary protein molecules come together to yield **<u>large aggregate structures</u>** with individual polypeptide chains known as **<u>sub-unit.</u>** **Haemoglobin**, a protein found in red blood cells. It consists of 4 polypeptide sub-units: <u>2 **α-sub-units**</u> and <u>2 **β-sub-units**</u>, each being **non-covalently** linked to a **haem** residue.
1(b)(i)	
1(b)(ii)	The presence of the valine residue in Hb-S causes **<u>Van der Waals' forces of attraction</u>** to be present, causing aggregation.
1(b)(iii)	At normal blood pH, which is about **<u>slightly basic</u>**, glutamic acid residue will be present as **<u>negatively charged glutamate</u>**, thus resulting in **<u>repulsion</u>** between the normal haemoglobin molecules.
1(c)(i)	$K_C = \frac{[Hb(O_2)_4]}{[Hb][O_2]^4}$ Units: $mol^{-4}\ dm^{12}$
1(c)(ii)	Since [Hb] = [Hb(O_2)$_4$] $K_C = \frac{1}{[O_2]^4}$ $K_C = \frac{1}{\left(7.6\times10^{-6}\right)^4} = 3.00\times10^{20}\,mol^{-4}dm^{12}$
1(c)(iii)	At equilibrium, Since 99[Hb] = [Hb(O_2)$_4$] $K_C = \frac{99}{[O_2]^4}$ $3.00\times10^{20} = \frac{99}{\left[O_2\right]^4_{eqm}}$ $[O_2]_{eqm}$ = 2.40×10^{-5} mol dm^{-3}
1(d)	$K_C = \frac{[Mb(O_2)]}{[Mb][O_2]}$ $1\times10^6 = \frac{[Mb(O_2)]}{[Mb]\left(7.6\times10^{-6}\right)}$ $\frac{[Mb(O_2)]}{[Mb]} = 7.6$

In ratio form, $[Mb(O_2)] : [Mb]$
7.6 : 1

% Mb in Mb-MbO2 mixture = $\dfrac{[Mb(O_2)]}{[Mb]+[Mb(O_2)]} X 100\% = \dfrac{7.6}{7.6+1} X 100\% = 88.4\%$

Alternatively,
Let the $[MbO_2]$ be x mol dm^{-3}

	Mb(aq)	+	O_2 (aq)	⇌	MbO_2 (aq)
Initial [ ]	1				0
Change	– x				+ x
Eqm [ ]	1 – x		7.6×10^{-6}		x

$$K_C = \frac{[Mb(O_2)]}{[Mb][O_2]}$$

$$1\times10^{6} = \frac{x}{(1-x)\left(7.6\times10^{-6}\right)}$$

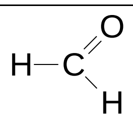
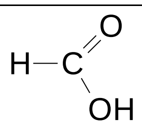
$$7.6 = \frac{x}{(1-x)}$$

x = 0.8837
% MbO_2 = 88.4%

2(a)(i)

Let oxidation number of C be x.
x + 4(+1) + (–2) = 0
x = –2

2(a)(ii)

A	B	C
	 **OR** carbon monoxide	O=C=O **OR** carbonic acid
O.N = 0	O.N = +2	O.N = +4

2(a)(iii)

A: $K_2Cr_2O_7$ / H_2SO_4(aq), heat with distillation
B: $K_2Cr_2O_7$ / H_2SO_4(aq), reflux
C: $KMnO_4$ / H_2SO_4(aq), reflux OR excess oxygen, heat

If students chooses CO as compound B and C, then the reagents and conditions will be: limited amount of oxygen, heat.

2(b)(i)

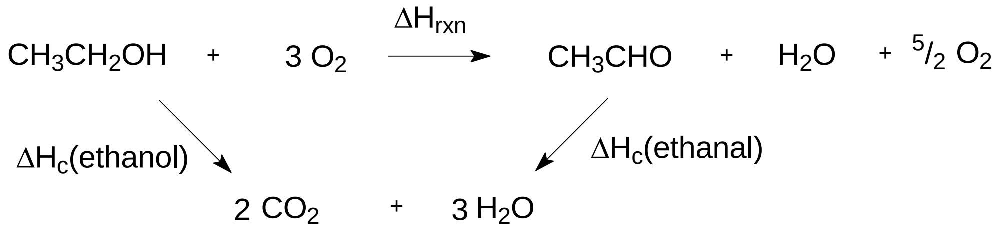
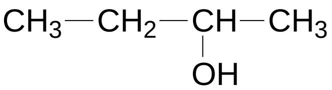
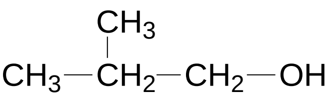
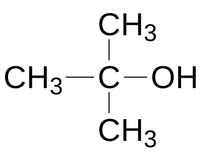
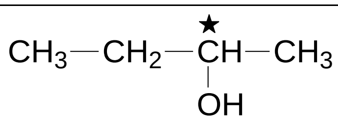
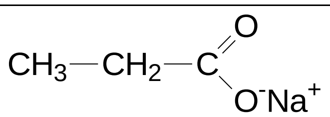
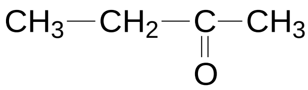
Q = mc□T
= (200)(4.18)(30)
= 25080 J

$n_{ethanol}$ = 1.50 / 46
= 0.03261 mol

Assuming 100% heating efficiency,

$$\Delta H_C = \frac{Q}{n_{ethanol}}$$

$$\Delta H_C = -\frac{25080}{0.03261} = -769 kJ\ mol^{-1}$$

2(b)(ii)	Heat is lost to the surroundings and to the copper can during the heating process.
2(c)(i)	$\Delta H_{rxn} = \left[(C-C)+5(C-H)+(C-O)+(O-H)+\frac{1}{2}(O=O)\right]$ $-\left[(C-C)+4(C-H)+(C=O)+2(O-H)\right]$ ΔH_{rxn} = [350 + 5(410) + 360 + 460 + ½ (496)] – [350 + 4(410) + 740 + 2(460)] = **<u>–182 kJ mol^{-1}</u>** If student is adopting the concept of bond breaking minus bond forming: $\Delta H_{rxn} = \left[(C-H)+(C-O)+\frac{1}{2}(O=O)\right]-\left[(C=O)+(O-H)\right]$ ΔH_{rxn} = [410 + 360 + ½ (496)] – [740 + 460] = **<u>–182 kJ mol^{-1}</u>**
2(c)(ii)	 ΔH_{rxn} = ΔH_c(ethanol) – ΔH_c(ethanal) ΔH_c(ethanal) = –1367 – (–182) = **<u>–1185 kJ mol^{-1}</u>**
2(c)(iii)	ΔH_c(ethanal) should be **<u>less exothermic</u>**. The combustion of ethanoic acid requires the **<u>breaking of the (strong) C-O</u>** and **<u>O-H bonds</u>**, that do not occur in ethanol. **OR** As ethanal is **<u>more oxidised</u>** than ethanol, and so its ΔH_c is **<u>less exothermic</u>**, therefore the even **<u>more oxidised ethanoic acid</u>** will have an **<u>even less exothermic</u>** ΔH_c.
2(d)(i)	**D:** $CH_3-CH_2-CH_2-CH_2-OH$  **E:**  **F:**  **G:**
2(d)(ii)	**D:** primary alcohol **E:** secondary alcohol **F:** primary alcohol **G:** tertiary alcohol
2(d)(iii)	 Compound E.
2(d)(iv)	 Compound E. and CH_3I
2(d)(v)	 Compound E.

<table>
<tr><td>3(a)</td><td>$2Mg(s) + O_2(g) \rightarrow 2MgO(s)$
Silvery Mg ribbon burns very vigorously and brightly with a bright white flame in oxygen to form a white residue

$P_4(s) + 5O_2(g) \rightarrow P_4O_{10}(s)$
P burns vigorously in excess oxygen with a bright yellow flame to form solid P_4O_{10}, which is a white residue

$S(s) + O_2(g) \rightarrow SO_2(g)$
Yellow solid S is burned in oxygen with a blue flame to form colourless gaseous SO_2 is formed, which will turn blue litmus paper red.</td></tr>
<tr><td>3(b)</td><td>$P_4O_{10}(s) + 6H_2O(l) \rightarrow 4H_3PO_4(aq)$

$SO_2(g) + H_2O(l) \rightarrow H_2SO_3(aq)$

Both oxides react readily with water to form acidic solution. Universal indicator will turn from green to red for both resulting solutions.</td></tr>
<tr><td>3(c)(i)</td><td>Amount of NaOH in 25 cm^3 = Amount of HCl
$= \frac{22.50}{1000} \times 0.100$
$= 2.250 \times 10^{-3}$ mol
Amount of NaOH in 100 cm^3 $= 2.250 \times 10^{-3} \times 4$
= <u>9.00 x 10^{-3} mol</u></td></tr>
<tr><td>3(c)(ii)</td><td>Amount of H_2O_2 in 25 cm^3 $= \frac{10.00}{1000} \times 0.0200 \times \frac{5}{2}$
$= 5.000 \times 10^{-4}$ mol
Amount of H_2O_2 in 100 cm^3 $= 5.000 \times 10^{-4} \times 4$
= <u>2.00 x 10^{-3} mol</u></td></tr>
<tr><td>3(c)(iii)</td><td>Amount of Na_2O_2 formed = Amount of H_2O_2 in 100 cm^3
= <u>2.00 x 10^{-3} mol</u>

Amount of NaOH formed from Na_2O_2 $= 2.00 \times 10^{-3} \times 2$
$= 4.000 \times 10^{-3}$ mol

Amount of NaOH formed from Na_2O $= 9.00 \times 10^{-3} - 4.00 \times 10^{-3}$
$= 5.000 \times 10^{-3}$ mol

Amount of Na_2O formed from Na_2O $= 5.00 \times 10^{-3} / 2$
= <u>2.50 x 10^{-3} mol</u></td></tr>
<tr><td>3(d)(i)</td><td>$Rate \quad \alpha \quad \frac{1}{time}$</td></tr>
<tr><td>3(d)(ii)</td><td>
<table>
<tr><td>Experiment</td><td>Time / s</td><td>Initial Rate / s^{-1}</td></tr>
<tr><td>1</td><td>33</td><td>0.0303</td></tr>
<tr><td>2</td><td>100</td><td>0.0100</td></tr>
<tr><td>3</td><td>67</td><td>0.0149</td></tr>
<tr><td>4</td><td>50</td><td>0.0200</td></tr>
</table>

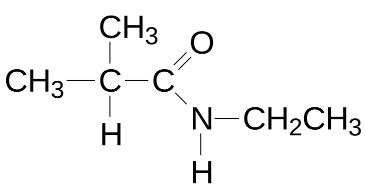
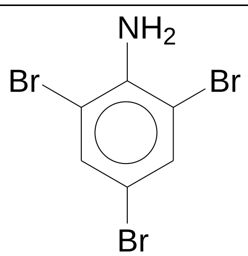
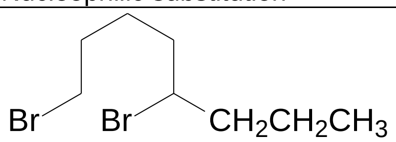
Since the total volume for all experiments are kept constant, $V_{reactant}$ α [reactant]

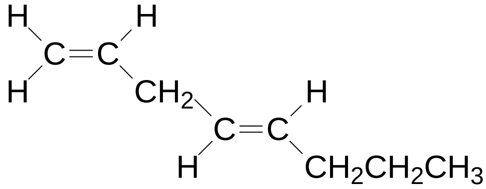
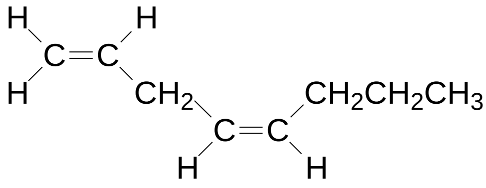
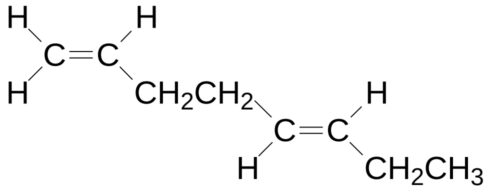
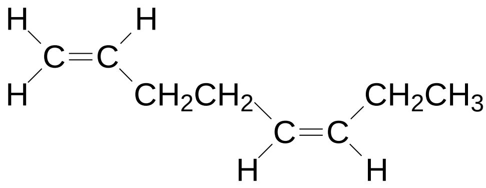
Compare reaction 2 and 3,
When volume of KI <u>increases by 50 %</u> and the rest of the reactants are kept constant, the rate of reaction <u>increases by 50%</u>.
Thus the <u>order of reaction with respect to KI is 1</u>.

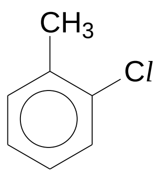
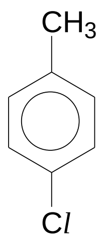
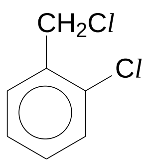
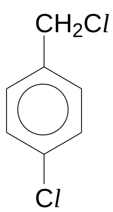
Compare reaction 2 and 4,
When volume of H_2O_2 <u>doubles</u> and the rest of the reactants are kept constant, the rate of reaction <u>doubles</u>
Thus the <u>order of reaction with respect to H_2O_2 is 1</u>.

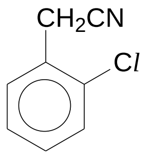
Compare reaction 1 and 2,
When volume of HCl <u>doubles</u> and the volume of the H_2O_2 <u>dropped to 1/3 of original value</u>, the rate of reaction <u>dropped to 1/3 of original value</u>.
Thus the <u>order of reaction with respect to HCl is 0</u>.</td></tr>
</table>

3(d)(iii)	**<u>Rate = k[KI][H_2O_2]</u>** Units of k = **<u>mol^{-1} dm^3 s^{-1}</u>** For the units, should students use the rate equation relationship in **(ii)**, then the units of k should be **<u>mol^{-2} dm^6 s^{-1}</u>** (this is the **<u>expected</u>** answer)

4(a)	1. The **molecules** of an ideal gas take up zero (or negligible) volume 2. The intermolecular **collisions** which occur between the ideal gas particles are **perfectly elastic** 3. The forces of attraction between the ideal gas particles (intermolecular attraction) as well as between the ideal gas particles and walls of the container are **negligible**
4(b)(i)	Methane is a simple molecular structure with intermolecular van der Waals' forces of attraction. Ammonia is a simple molecular structure with intermolecular hydrogen bonding. More energy is required to overcome the **<u>stronger intermolecular hydrogen bonding</u>** in ammonia as compared to the weaker intermolecular van der Waals' forces of attraction in methane. Hence, ammonia has a higher boiling point.
4(b)(ii)	When methane is liquefied, the molecules in liquid state will come **<u>closer together</u>**. Hence, the arrangement of the molecules will become **<u>more ordered</u>** as compared to the gaseous state. Hence, **<u>entropy change</u>** will decrease/become **<u>negative</u>**.
4(b)(iii)	□G = □H – T□S 0 = 23300 – (–33 + 273.15)□□S □S = **<u>+97.0 J mol^{-1} K^{-1}</u>**
4(c)(i)	Order of basic strength: ethylamine > ammonia > phenylamine The **<u>electron donating ethyl group</u>** increases the **<u>availability of the lone pair of electrons on N atom</u>**. The N atom in ethylamine is more available to accept a proton via a dative bond than that in ammonia. The electron donating ethyl group **<u>disperses the positive charge</u>** on the N atom, thus **<u>stabilising the conjugate acid</u>** relative to the base. Hence ethylamine is a stronger base than ammonia. The **<u>lone pair of electrons on the N atom</u>** in phenylamine is **<u>delocalised into the π electron cloud in benzene</u>**. The N atom in phenylamine is **<u>less available</u>** to accept a proton via a dative bond than that in ammonia. The positive charge on the N atom is not dispersed through delocalization, thus **<u>no stabilisation of conjugate acid</u>** relative to base. Hence, phenylamine is a weaker base than ammonia and ethylamine.
4(c)(ii)	$K_b = \frac{[C_2H_5NH_3^+][OH^-]}{[C_2H_5NH_2]}$ Let [OH^-] be x. $6.4\times10^{-4} = \frac{x^2}{0.100}$ x = 0.008 mol dm^{-3} pOH = –log[OH^-] = 2.10 pH = 14 – pOH = **<u>11.9</u>**
4(c)(iii)	 
4(d)(i)	Nucleophilic substitution
4(d)(ii)	 1,5-dibromooctane

4(d)(iii)	   

5(a)	Volatility **<u>decreases</u>** down the group and the intensity of colour **<u>increases</u>** down the group. Halogens exist as **<u>simple molecular</u>** structures with **<u>intermolecular</u>** weak **<u>Van der Waals</u>**' forces of attraction. Down the group, molecular size increases (**<u>number of electrons increase</u>**). Van der Waals' forces become **<u>more extensive</u>** Boiling point increases, thus volatility decreases. Down the group, **<u>fluorine</u>** is **<u>pale yellow gas</u>**, **<u>chlorine</u>** is **<u>yellowish-green gas</u>**, **<u>bromine</u>** is **<u>reddish brown liquid</u>** and **<u>iodine</u>** is **<u>shiny black crystals</u>**.
5(b)	NaC*l* (aq) + H_2SO_4 (*l*) → $NaHSO_4$ (aq) + HC*l* (g) (No redox reaction) White fumes of HC*l observed* NaBr (aq) + H_2SO_4 (*l*) → $NaHSO_4$ (aq) + HBr (g) 2 HBr (g) + H_2SO_4 (*l*) → Br_2 (g) + $2H_2O$ (*l*) + SO_2 (g) White fumes of HBr. Reddish-brown fumes of Br_2 gas. Pungent SO_2 gas NaI (aq) + H_2SO_4 (*l*) → $NaHSO_4$ (aq) + H I (g) 2 HI (g) + H_2SO_4 (*l*) → I_2 (g) + $2H_2O$ (*l*) + SO_2 (g) 6HI (g) + H_2SO_4 (*l*) → 3 I_2 (g) + 4 H_2O (*l*) + S (s) 8HI (g) + H_2SO_4 (*l*) → 4 I_2 (g) + H_2S (g) + $4H_2O$ (*l*) White fumes of HI. Violet fumes of I_2 gas. Pungent SO_2 gas. Yellow S solid seen "Rotten egg" smell, H_2S gas
5(c)	Electrophilic substitution occurred to produce **K**.  or  Free radical substitution on **K** to form **L**.  or  Nucleophilic substitution on **L** to form **M**.



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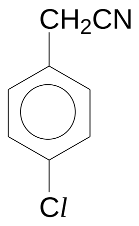
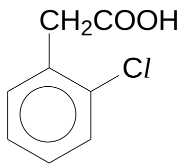


Acidic hydrolysis occurred on **M** to form **N**.



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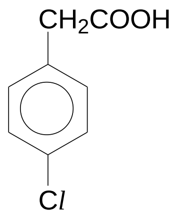
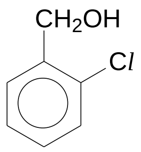
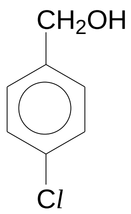


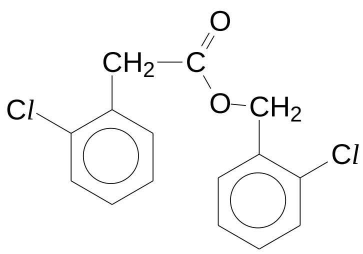
Nucleophilic substitution on **L** to form **P**.



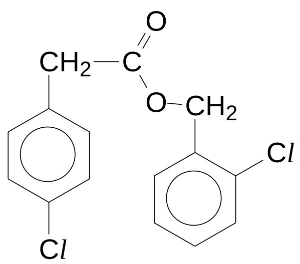
or



Esterification occurred with **N** and **P** to form **Q.**



or



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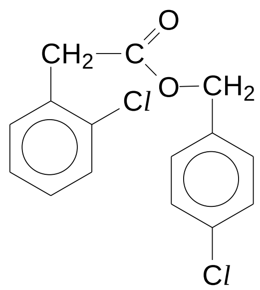
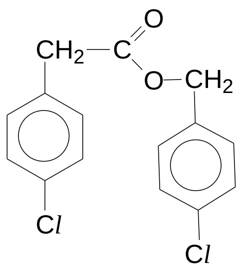


or



END